GlucoTwin AI: Time-Series Digital Twin for Glucose Forecasting

Unstop Digital Twin Challenge 2026 | Comprehensive Proof of Concept

- **Project Mission:** Develop an educational physiological digital twin predicting glucose trends 2 hours ahead.
- **Core Paradigm:** Continuously updates virtual patient state using time-series sensor data & contextual events.
- **Honest Science:** Compares trained ML models against simple persistence baselines with zero future data leakage.
- **Regulatory Status:** Purely educational and research demonstration. Not a medical device. Non-diagnostic.

Clinical Need & The Glycemic Forecasting Problem

Beyond Reactive Continuous Glucose Monitoring

- **Glycemic Volatility:** Diabetes affects over 530 million adults globally; postprandial spikes and nocturnal hypoglycemia pose acute risks.
- **Lagged Observation:** Standard CGMs present reactive snapshots with 10-15 minute interstitial physiological delays.
- **The Need for Forward Horizon:** A 2-hour (120-minute) predictive horizon allows anticipatory lifestyle adjustments.
- **Digital Twin Potential:** A software twin maintains memory of recent carbohydrates, exertion, and circadian rhythms.

The GlucoTwin AI Solution Concept

A Closed-Loop Virtual Metabolic Avatar

- **Virtual Twin Synchronization:** Ingests 5-minute telemetry ticks into an in-memory sliding window state buffer.
- **Multi-Modal Context Integration:** Blends glucose values with heart rate, physical activity index, meal carbs, and sleep.
- **Defensible Uncertainty:** Inter-tree variance delivers calibrated 95% confidence bounds rather than fabricated certainty.
- **Transparent Simulation Controls:** Provides real-time streaming, pause, step (+5m/+30m), reset, and what-if sandbox.

Healthcare Use Case & Regulatory Non-Diagnostic Scope

Safety-by-Design & Clear Boundaries

- **Target Use Case:** Research testbed, metabolic education, and simulation environment for digital health researchers.
- **Strict Safety Boundaries:** Does NOT provide insulin dose titration, medical diagnosis, or emergency intervention.
- **Regulatory Scope:** Clarified as non-SaMD (Software as a Medical Device) under FDA 21 CFR 860 and EU MDR Annex VIII.
- **Patient Privacy:** Zero collection of Protected Health Information (PHI); utilizes strictly synthetic, de-identified identities.

End-to-End Four-Tier System Architecture

From Deterministic Physics to Interactive Dashboard

- **Tier 1 - Telemetry Engine:** Deterministic synthetic generator modeling digestion curves and homeostatic drift.
- **Tier 2 - Feature Engineering:** 23 backward-looking features including multi-step lags, rolling statistics, and harmonics.
- **Tier 3 - ML Ensemble:** 150-tree Random Forest optimized for 24-step (120 min) horizon with uncertainty estimation.
- **Tier 4 - Delivery:** FastAPI REST services (/api/v1/...) paired with React 18, TypeScript, Tailwind CSS, and Recharts.

Synthetic Physiological Dataset & Generation Dynamics

Biologically Grounded Time-Series Synthesis

- **Sampling Frequency:** Exactly 5-minute ticks across 5,500 sequential observations (multi-week synthetic trial).
- **Gastric Emptying Delay:** Exponential convolution of meal carbohydrate events simulating gradual digestive absorption.
- **Activity & Uptake:** Physical exertion modulates insulin sensitivity and accelerates muscular glucose uptake.
- **Heart Rate & Circadian Dynamics:** Correlated cardiovascular response and sinusoidal diurnal baseline shifts.

Feature Engineering & Zero Data Leakage Guarantee

Rigorous Time-Series Feature Construction

- **Chronological Sequence:** Temporal sorting enforced; past observations strictly segregated from future targets.
- **Lag Operators:** Lag 1 (5m), Lag 2 (10m), Lag 3 (15m), Lag 6 (30m), Lag 12 (1h), and Lag 24 (2h).
- **First Derivatives:** Glycemic velocity calculated over 15-minute and 30-minute backward intervals (mg/dL/min).
- **Cyclical Harmonic Encoding:** Sine and cosine hour representations preserve continuity across midnight boundary.

Machine Learning Ensemble & Uncertainty Quantification

Ensemble Variance for Defensible Confidence Intervals

- **Model Architecture:** RandomForestRegressor (150 estimators, max_depth=14, min_samples_split=4).
- **2-Hour Target Formulation:** Precisely y(t+24), forecasting glucose exactly 120 minutes ahead.
- **Defensible Uncertainty:** Computes empirical variance across all 150 individual decision tree estimates.
- **95% Prediction Interval:** [y_hat - 1.96*sigma, y_hat + 1.96*sigma] dynamically rendered on telemetry charts.

Empirical Evaluation vs. Persistence Baseline

Honest Metric Reporting on 1,091 Chronological Test Samples

- **Trained Model MAE:** 5.260 mg/dL (vs. Persistence: 48.711 mg/dL — **+89.2% improvement**).
- **Trained Model RMSE:** 7.160 mg/dL (vs. Persistence: 59.482 mg/dL — **+88.0% improvement**).
- **Why Persistence Fails at 2h:** Glycemic states experience substantial movement over 120m due to meals and activity.
- **Top Feature Importance:** Circadian harmonics (cos_hour: 56.2%), current glucose (11.5%), and lag-1 (8.4%) dominate.

Interactive What-If Sandbox, Limitations & Future Scope

Next Steps for Digital Twin Research

- **What-If Sandbox:** Sliders for meal carbs and physical activity project immediate hypothetical 2-hour deviations.
- **Identified Limitations:** Evaluated on synthetic sequences; real patients exhibit variable insulin sensitivity and illness.
- **Clinical Translation Roadmap:** Ingest licensed real-world datasets (OhioT1DM / D1NAMO) and add LSTM/Transformer models.
- **Summary:** Fully runnable, documented, test-verified end-to-end digital twin prototype for Unstop 2026.